Design Fundamentals 3/4

Topics 13–18

13. Specialized storage paradigms

- Blob storage (S3/GCS), time-series DBs (InfluxDB, Prometheus), graph DBs (Neo4j), spatial indexes (quadtrees, geohash)

14. Replication and sharding

- Sync vs async replicas
- Sharding by range, hash or region
- Hot spots
- Resharding with consistent hashing

15. Leader election

- One leader handles writes/jobs
- Consensus via Paxos/Raft
- In practice use ZooKeeper/etcd leases

16. Peer-to-peer networks

- Split a large file into chunks shared between peers (BitTorrent)
- Gossip protocols
- DHT for peer discovery

17. Polling and streaming

- Polling at intervals vs a long-lived connection (WebSockets/SSE)
- Streaming suits chat and live data

18. Configuration

- Static (deployed with the code, reviewed) vs dynamic (a config service, instant but riskier)

Recap

- 13. Specialized storage paradigms
- 14. Replication and sharding
- 15. Leader election
- 16. Peer-to-peer networks
- 17. Polling and streaming
- 18. Configuration